

The Fabius Function and Rvachev's Up Function

Smoothness, exact dyadic arithmetic, Fourier structure,
and sharp endpoint asymptotics

A human-readable exposition based on the complete Lean development
in `VladimirReshetnikov/ProveIt`

This article synthesizes the formal development in `Analysis/FabiusFunction` on the repository's main branch as inspected in August 2026. It distinguishes the bounded cumulative-distribution version of the Fabius function from its signed global extension, and it states conjectural material as conjectural. The proofs below are ordinary mathematical proofs; the Lean source provides a machine-checked refinement of the same dependency structure.

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Abstract

The Fabius function is a canonical smooth transition from 0 to 1 whose self-similarity is governed by the differential equation $F'(x) = 2F(2x)$. Its even twin, Rvachev's compactly supported function up , is the unique normalized smooth solution of

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)), \quad \text{supp up} = [-1, 1], \quad \text{up}(0) = 1.$$

These functions simultaneously belong to probability, harmonic analysis, combinatorics, exact arithmetic, approximation theory, and asymptotic analysis. The present article develops their equivalent constructions, existence and uniqueness, smoothness and failure of analyticity on the support, Fourier products, Poisson summation, moments, polynomial and Legendre expansions, and the signed global extension controlled by the Thue–Morse sequence.

Particular attention is paid to two topics. First, every dyadic value is rational. We prove both a closed finite Thue–Morse formula and a recursive Taylor-block formula that removes the highest binary digit of the numerator. These formulas lead to exact algorithms, denominator bounds, 2-adic valuations, the integral and odd Reshetnikov numbers, and finite q -binomial identities. Second, we derive the sharp small- x expansion of $\log F(x)$. Its natural coordinate is the lower Lambert phase

$$\lambda(x) = -\frac{W_{-1}(-x \log 2)}{\log 2}, \quad \lambda(x)2^{-\lambda(x)} = x,$$

not merely $-\log_2 x$. The expansion contains an explicit, nonconstant, smooth one-periodic term sampled at $\lambda(x)$; omitting this term makes the often quoted $O(1/\log x)$ remainder false. We give the exact periodic Laplace decomposition, its Gamma–zeta Fourier coefficients and mean, the corrected Lambert form, an elementary logarithmic form, the first periodic saddle correction, and the full all-orders expansion in inverse powers of $\lambda(x)$.

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1 Orientation: three functions, not one

The notation surrounding the Fabius function is historically inconsistent. One source writes a globally defined signed function as F ; another reserves F for the increasing function on $[0, 1]$; still another begins with the compact bump up. Mixing these conventions is harmless for some identities and disastrous for others. We therefore fix the following three objects from the outset.

Definition 1.1 (The bounded Fabius function). The bounded Fabius function is the unique map $F: \mathbb{R} \rightarrow [0, 1]$ satisfying

$$F(x) = 0 \quad (x \leq 0), \quad (1)$$

$$F(x) = 1 \quad (x \geq 1), \quad (2)$$

$$F(1-x) = 1 - F(x) \quad (0 \leq x \leq 1), \quad (3)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (4)$$

and $F \in C^\infty(\mathbb{R})$. We use the clamped extension in (1)–(2), even when only the restriction to $[0, 1]$ is under discussion.

Definition 1.2 (Rvachev's up-function). The up-function associated with F is

$$\text{up}(x) = \begin{cases} F(x+1), & x \leq 0, \\ F(1-x), & x > 0. \end{cases} \quad (5)$$

Thus up is supported in $[-1, 1]$, and on that interval

$$F(x) = \text{up}(x-1) \quad (0 \leq x \leq 1), \quad \text{up}(x) = 1 - F(x) \quad (0 \leq x \leq 1). \quad (6)$$

Let $w_2(n)$ denote the number of 1's in the binary expansion of n , and put

$$\varepsilon_n = (-1)^{w_2(n)}. \quad (7)$$

The elementary identities

$$\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n \quad (8)$$

are the algebraic engine behind the global self-similarity.

Definition 1.3 (The signed global extension). Define

$$\Theta(x) = \sum_{n=0}^{\infty} \varepsilon_n \text{up}(x - 2n - 1). \quad (9)$$

The sum is locally finite, because the supports of the summands have disjoint interiors. On the band $[2m, 2m+2]$,

$$\Theta(x) = \varepsilon_m \text{up}(x - 2m - 1). \quad (10)$$

In particular, $\Theta = F$ on $[0, 1]$ and $\Theta(x) = 0$ for $x \leq 0$. Unlike the bounded F , however, Θ is signed and nonconstant beyond 1.

We will usually write F for the bounded function and Θ for the signed extension. Statements from older literature that use F globally are to be read with F replaced by Θ .

1.1 Historical coordinates

The same distribution or function appeared independently in several guises: in work of Jessen and Wintner on infinite convolutions, in Fabius's probabilistic example of a nowhere analytic C^∞ function, in Rvachev's compactly supported solutions of functional-differential equations, and later in work of Kirov–Totkov, Arias de Reyna, Schnabl, and others. Arias de Reyna's 1982 paper gives a systematic construction and a large collection of identities; his later paper develops the arithmetic of the dyadic values. The modern formal development on which this article is based proves the results of both papers, supplies missing analytic bridges, corrects several endpoint and normalization defects, and extends the asymptotic theory substantially.

The most useful conceptual slogan is this:

One object, four viewpoints

The bounded F is a cumulative distribution function; up is the density of a centered version of the same random series; $\widehat{\text{up}}$ is an infinite sinc product; and Θ is the Thue–Morse signed global continuation satisfying a pure dilation differential equation.

2 Probability and infinite convolution

2.1 The random series

Let $(U_n)_{n \geq 1}$ be independent random variables, each uniformly distributed on $[0, 1]$, and set

$$X = \sum_{n=1}^{\infty} \frac{U_n}{2^n}. \quad (11)$$

The series converges almost surely and in every L^p , and $0 \leq X \leq 1$. Its law is symmetric about $1/2$, since replacing each U_n by $1 - U_n$ changes X into $1 - X$ without changing the joint law.

It is often cleaner to center and rescale:

$$Y = 2X - 1 = \sum_{j=0}^{\infty} V_j, \quad V_j \sim \text{Unif}[-2^{-j-1}, 2^{-j-1}], \quad (12)$$

with the V_j independent. The support widths sum to 1, so $Y \in [-1, 1]$.

Theorem 2.1 (Probabilistic realization). *The random variable X has cumulative distribution function F , and $Y = 2X - 1$ has density up . Equivalently,*

$$F(x) = \mathbb{P}(X \leq x), \quad 0 \leq x \leq 1, \quad (13)$$

and

$$\mathbb{P}(Y \in A) = \int_A \text{up}(t) dt \quad (14)$$

for every Borel set $A \subseteq \mathbb{R}$.

Proof. For a finite truncation $Y_N = \sum_{j=0}^N V_j$, the law is the convolution of the uniform measures on the indicated intervals. Its characteristic function, in the Fourier convention $\widehat{g}(z) = \int g(t) e^{-2\pi i t z} dt$, is

$$\prod_{j=0}^N \frac{\sin(\pi z / 2^j)}{\pi z / 2^j}. \quad (15)$$

The random variables Y_N converge almost surely to Y , so their laws converge weakly. In [section 5](#) we prove that the limiting product is the Fourier transform of a C^∞ density supported in $[-1, 1]$; call that density u . Consequently Y has density u .

Let H be the distribution function of $X = (Y + 1)/2$. On $-1 \leq y \leq 0$,

$$u(y) = \frac{1}{2} H' \left(\frac{y+1}{2} \right).$$

The refinement identity proved below implies $\frac{1}{2} H'((y+1)/2) = H(y+1)$. Hence $u(y) = H(y+1)$ on the left half of the support. Symmetry gives $u(y) = H(1-y)$ on the right half. Thus H and u satisfy exactly the bridge in (5). The uniqueness theorem in [section 3](#) identifies $H = F$ and $u = \text{up}$. $\square$

The theorem also gives the left-half identity that appeared in the original probabilistic construction:

$$\text{up}(x) = \mathbb{P} \left(\sum_{n=1}^{\infty} \frac{U_n}{2^n} \leq x+1 \right), \quad -1 \leq x \leq 0. \quad (16)$$

The apparent mismatch between “ F is a CDF” and “ up is a density” is not a contradiction: the differential self-similarity turns the left half of the density of Y into the CDF of X .

2.2 The refinement and delay-differential equations

Separate the first summand in (12):

$$Y = V_0 + \frac{1}{2} Y', \quad (17)$$

where $V_0 \sim \text{Unif}[-1/2, 1/2]$ and Y' is an independent copy of Y . The density of $Y'/2$ is $2 \text{up}(2x)$. Convolution with the density $\mathbf{1}_{[-1/2, 1/2]}$ gives the integral refinement law

$$\text{up}(x) = \int_{-1/2}^{1/2} 2 \text{up}(2(x-v)) dv = \int_{2x-1}^{2x+1} \text{up}(t) dt. \quad (18)$$

Differentiating yields Rvachev's equation.

Theorem 2.2 (Rvachev equation). *For every $x \in \mathbb{R}$,*

$$\text{up}'(x) = 2(\text{up}(2x+1) - \text{up}(2x-1)). \quad (19)$$

Moreover up is even, nonnegative, supported in $[-1, 1]$, normalized by

$$\text{up}(0) = 1, \quad \int_{\mathbb{R}} \text{up}(x) dx = 1. \quad (20)$$

Proof. Equation (19) follows from the fundamental theorem of calculus applied to (18). The distribution of Y is symmetric, hence up is even. Its nonzero set is contained in $[-1, 1]$ because the sum of the half-widths in (12) is 1. Its closed (topological) support is exactly $[-1, 1]$: every open subinterval of $(-1, 1)$ receives positive mass from a suitable finite convolution, while the remaining tail can be confined to an arbitrarily small interval with positive probability. Normalization of the integral is the normalization of a probability density.

Finally, put $x = 0$ in (18). The integral is the total mass on $[-1, 1]$, so $\text{up}(0) = 1$. $\square$

Corollary 2.3 (The local Fabius equation). *The function $F(x) = \text{up}(x - 1)$ on $[0, 1]$ satisfies*

$$F'(x) = 2F(2x), \quad 0 \leq x \leq \frac{1}{2}. \quad (21)$$

It also satisfies $F(1 - x) = 1 - F(x)$.

Proof. For $0 \leq x \leq 1/2$, use (19) at $x - 1$:

$$F'(x) = \text{up}'(x - 1) = 2(\text{up}(2x - 1) - \text{up}(2x - 3)) = 2\text{up}(2x - 1) = 2F(2x),$$

because $2x - 3 \leq -2$ lies outside the support. Symmetry of X about $1/2$ and continuity of its law imply $F(1 - x) = \mathbb{P}(X \leq 1 - x) = \mathbb{P}(X \geq x) = 1 - F(x)$. $\square$

2.3 Finite convolutions and constructive approximants

Let

$$\rho_j(x) = 2^j \mathbf{1}_{[-2^{-j-1}, 2^{-j-1}]}(x), \quad u_N = \rho_0 * \rho_1 * \cdots * \rho_N. \quad (22)$$

Then u_N is a compactly supported piecewise polynomial density, and

$$\widehat{u_N}(z) = \prod_{j=0}^N \text{sinc}\left(\frac{\pi z}{2^j}\right), \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases} \quad (23)$$

For every fixed m , the products converge in the weighted L^1 norm $\int (1 + |z|)^m |\cdot| dz$. Fourier inversion therefore gives

$$u_N \longrightarrow \text{up} \quad \text{in } C^m(\mathbb{R}) \quad (24)$$

for every m . This is an explicit constructive existence proof, and it also explains why increasingly many convolutions produce an increasingly flat bump.

A related discrete approximation starts with the polynomials

$$p_0(x) = 1, \quad p_{m+1}(x) = p_m(x)(1 + x + \cdots + x^{2^{m+1}-1}). \quad (25)$$

The coefficient of x^j in $\prod_{r=1}^m (1 + x + \cdots + x^{2^r-1})$ counts restricted digit sums and supplies the height of a dyadic step approximant. With half-weights at cell endpoints, these step functions converge pointwise to up ; without the half-weights the endpoint statement is false. The weak convergence is simply the convergence of the finite random sums, while continuity of the limiting density upgrades it to the stated pointwise limit.

3 Existence, uniqueness, monotonicity, and support

3.1 Rapid decay of the sinc product

For $z \in \mathbb{C}$ define

$$\Phi(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^j}\right). \quad (26)$$

Since $1 - \text{sinc } w = O(w^2)$ as $w \rightarrow 0$, the product converges uniformly on compact subsets of $\mathbb{C}$ and defines an entire even function. On the real axis it decays faster than every power.

Lemma 3.1 (Schwartz decay). *For each integer $N \geq 0$ there is a constant C_N such that*

$$|\Phi(t)| \leq C_N (1 + |t|)^{-N}, \quad t \in \mathbb{R}. \quad (27)$$

The same is true after any fixed number of derivatives.

Proof. For real y ,

$$|\operatorname{sinc} y| \leq \min\{1, |y|^{-1}\}.$$

Fix N and retain the first N factors of (26). Their product is $O(|t|^{-N})$, while every remaining factor has absolute value at most 1. This proves (27) away from a bounded interval; continuity handles the interval. Derivatives are finite sums of products in which finitely many factors are differentiated. The differentiated sinc factors have the same type of polynomial bound, and one may retain additional undifferentiated factors to obtain arbitrarily high decay. $\square$

Consequently the inverse Fourier integral

$$u(x) = \int_{\mathbb{R}} \Phi(t) e^{2\pi i t x} dt \quad (28)$$

converges absolutely with all derivatives. It gives a Schwartz function. On the other hand, Φ is the characteristic function of the random series Y , whose law is supported in $[-1, 1]$. Hence u is the density of that law and vanishes outside $[-1, 1]$. This proves existence of a smooth compactly supported solution.

3.2 Fourier uniqueness

The Fourier transform turns the delay-differential equation into a one-step refinement equation.

Lemma 3.2 (Transform of Rvachev's equation). *Let $u \in C^1(\mathbb{R})$ have compact support and satisfy*

$$u'(x) = k(u(2x+1) - u(2x-1)).$$

Then

$$\widehat{u}(z) = \frac{k}{2} \operatorname{sinc}(\pi z) \widehat{u}(z/2). \quad (29)$$

If $\widehat{u}(0) \neq 0$, necessarily $k = 2$.

Proof. Integration by parts gives $\widehat{u}'(z) = 2\pi i z \widehat{u}(z)$. In the first shifted term put $y = 2x + 1$, and in the second put $y = 2x - 1$. The right-hand side transforms to

$$\frac{k}{2} (e^{\pi i z} - e^{-\pi i z}) \widehat{u}(z/2) = k i \sin(\pi z) \widehat{u}(z/2).$$

Division by $2\pi i z$, with the value at $z = 0$ filled in continuously, proves (29). Letting $z \rightarrow 0$ gives $\widehat{u}(0) = (k/2)\widehat{u}(0)$, hence $k = 2$ when the integral of u is nonzero. $\square$

For $k = 2$, iteration gives

$$\widehat{u}(z) = \left(\prod_{j=0}^{N-1} \operatorname{sinc}\left(\frac{\pi z}{2^j}\right) \right) \widehat{u}(z/2^N). \quad (30)$$

As $N \rightarrow \infty$, continuity at the origin yields

$$\widehat{u}(z) = \widehat{u}(0) \Phi(z). \quad (31)$$

Theorem 3.3 (Existence and uniqueness). *There is a unique C^∞ function $u: \mathbb{R} \rightarrow \mathbb{R}$ with compact support, $u(0) = 1$, and*

$$u'(x) = 2(u(2x + 1) - u(2x - 1)). \quad (32)$$

It is up. Equivalently, there is a unique bounded Fabius function in the sense of [theorem 1.1](#).

Proof. Existence was obtained from the inverse transform (28). Its integral is $\Phi(0) = 1$, and the refinement equation follows either by Fourier inversion or by conditioning as in [theorem 2.2](#). The value at zero is 1 by (18).

For uniqueness, let u be any other solution and write $U = \widehat{u}$. Compact support makes U entire, and the differential equation gives $U(z) = \text{sinc}(\pi z)U(z/2)$. We first show that $U(0) \neq 0$. Otherwise, if U were not identically zero, let $m \geq 1$ be the least index of a nonzero Taylor coefficient, say $U(z) = a_m z^m + O(z^{m+1})$. Since $\text{sinc}(\pi z) = 1 + O(z^2)$, comparison of the first nonzero terms in the recurrence would give $a_m = 2^{-m}a_m$, an impossibility. Hence $U \equiv 0$, so $u \equiv 0$ by Fourier uniqueness, contradicting $u(0) = 1$. Thus $U(0) \neq 0$, and iteration of the recurrence yields $U(z) = U(0)\Phi(z)$. Therefore u is a scalar multiple of the inverse transform of Φ ; evaluation at 0 fixes that scalar to be 1.

Given u , define F by (6) on $[0, 1]$ and clamp it outside. The support and endpoint flatness of u make the clamped function smooth, and [theorem 2.3](#) gives the Fabius equation. Conversely (5) recovers u from any bounded Fabius function, so uniqueness of u implies uniqueness of F . $\square$

Remark 3.4. The normalization $u(0) = 1$ is geometrically natural, while $\int u = 1$ is probabilistically natural. For a solution of (32) with compact support the two normalizations agree after scaling, and the canonical solution satisfies both.

3.3 Monotonicity and strict inequalities

Proposition 3.5 (Monotonicity). *The bounded Fabius function is nondecreasing on $\mathbb{R}$. More precisely,*

$$0 < F(x) < 1 \quad (0 < x < 1). \quad (33)$$

Consequently $\text{up}(x) > 0$ for $-1 < x < 1$.

Proof. On $[0, 1/2]$, equation (21) and nonnegativity give $F' \geq 0$. Symmetry transfers monotonicity to $[1/2, 1]$, and the clamped pieces are compatible.

For strict positivity, suppose $F(x) = 0$ at some $x > 0$. If $x \geq 1/2$, monotonicity gives $F(1/2) \leq F(x) = 0$, contradicting $F(1/2) = 1/2$. Hence $0 < x < 1/2$. Since $F \geq 0$, the point x is a global minimum, so $F'(x) = 0$. The differential equation then gives $F(2x) = 0$. Iterating this argument while the current point lies below $1/2$ eventually produces a zero at or to the right of $1/2$, which is already impossible. Thus $F(x) > 0$ for $x > 0$. Applying this to $1 - x > 0$ and using symmetry yields $F(x) < 1$ for $x < 1$. The bridge (5) gives strict positivity of up in the interior of its support. $\square$

Corollary 3.6 (Exact support). *The pointwise support (the set on which up is nonzero) is $(-1, 1)$, while the closed or topological support is $[-1, 1]$.*

3.4 A first picture

The following plots use exact dyadic values on a fine grid and merely join them by line segments. The underlying functions are, of course, smooth.

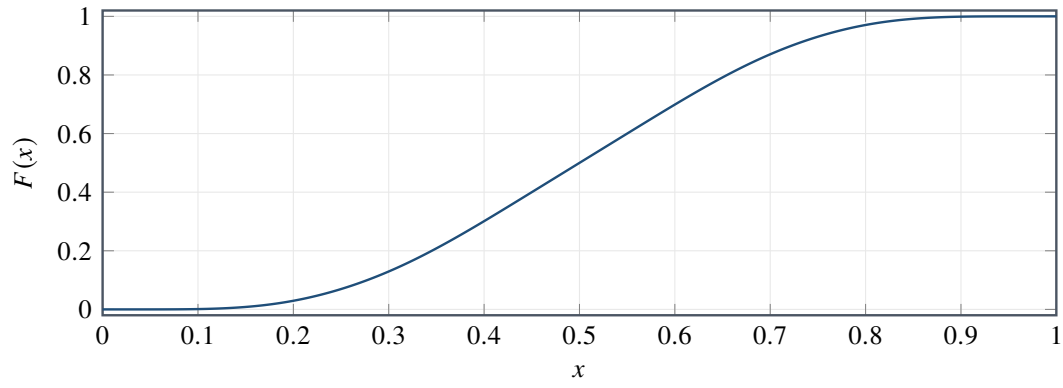


Figure 1: The bounded Fabius function on $[0, 1]$.

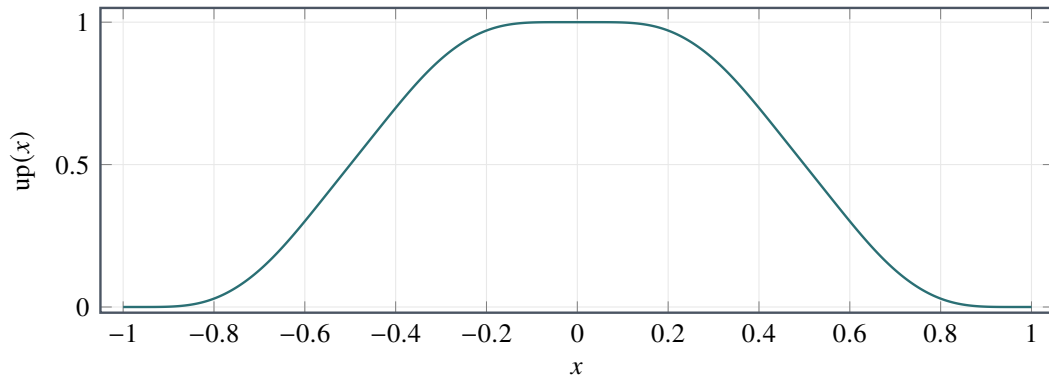


Figure 2: Rvachev's even compactly supported up-function.

4 The signed global extension and differential self-similarity

The signed extension Θ removes the artificial clamping at 1. Its advantage is that the local equation becomes a global one with no reflected term.

Theorem 4.1 (Global differential equation). *The function Θ is C^∞ on $\mathbb{R}$ and satisfies*

$$\Theta'(x) = 2\Theta(2x), \quad x \in \mathbb{R}. \quad (34)$$

For every $k \geq 0$,

$$\Theta^{(k)}(x) = 2^{\binom{k+1}{2}} \Theta(2^k x). \quad (35)$$

On the support of up ,

$$\text{up}^{(k)}(x) = 2^{\binom{k+1}{2}} \Theta(2^k(x+1)), \quad -1 \leq x \leq 1. \quad (36)$$

Proof. Because the sum (9) is locally finite, it may be differentiated term by term. Apply (19):

$$\begin{aligned} \Theta'(x) &= 2 \sum_{n \geq 0} \varepsilon_n [\text{up}(2x - 4n - 1) - \text{up}(2x - 4n - 3)] \\ &= 2 \sum_{n \geq 0} \varepsilon_{2n} \text{up}(2x - 2(2n) - 1) + 2 \sum_{n \geq 0} \varepsilon_{2n+1} \text{up}(2x - 2(2n+1) - 1) \\ &= 2\Theta(2x), \end{aligned}$$

where (8) was used in the second equality. Repeated differentiation proves (35): the factors accumulated are $2^1, 2^2, \dots, 2^k$, whose product is $2^{\binom{k+1}{2}}$. On $[-1, 1]$ one has $\text{up}(x) = \Theta(x+1)$; differentiating and applying (35) gives (36). $\square$

Corollary 4.2 (Flat endpoints). *Every derivative of F vanishes at 0, and every derivative of up vanishes at -1 and 1:*

$$F^{(k)}(0) = 0, \quad \text{up}^{(k)}(-1) = \text{up}^{(k)}(1) = 0 \quad (k \geq 0). \quad (37)$$

Proof. The clamped function is identically zero to the left of 0, so smoothness already forces the first statement. Alternatively, use (35) and $\Theta(0) = 0$. Equation (36) gives the value at -1 directly. Evenness supplies the value at 1. $\square$

The next phenomenon is one of the function's most striking features: at a dyadic point the Taylor series is a polynomial, but the function does not equal that polynomial in any neighborhood.

Theorem 4.3 (Dyadic jets). *Let*

$$x_{n,a} = \frac{a - 2^n}{2^n}, \quad 0 \leq a \leq 2^{n+1}. \quad (38)$$

Then

$$\text{up}^{(k)}(x_{n,a}) = 0 \quad (k > n). \quad (39)$$

If a is odd, then

$$\text{up}^{(n)}(x_{n,a}) \neq 0. \quad (40)$$

Hence the formal Taylor polynomial at a reduced level- n dyadic point has degree exactly n .

Proof. By (36),

$$\text{up}^{(k)}(x_{n,a}) = 2^{\binom{k+1}{2}} \Theta\left(2^k \frac{a}{2^n}\right).$$

If $k > n$, the argument is the even integer $2^{k-n}a$. Formula (10) shows that Θ vanishes at every even nonnegative integer, because these points are support endpoints of the translated bumps. This proves (39). When $k = n$ and a is odd, write $a = 2m + 1$. Then $\Theta(a) = \varepsilon_m \text{up}(0) = \varepsilon_m \neq 0$, proving (40). $\square$

Theorem 4.4 (Failure of analyticity on the support). *The function up is not real analytic at any point of $[-1, 1]$. Consequently F is not real analytic at any point of $[0, 1]$.*

Proof. Reduced dyadic points of arbitrarily high level are dense in $[-1, 1]$. Suppose up were analytic at some x in the support. It would then be analytic on a small open interval I around x . Choose a reduced dyadic $y = x_{n,a} \in I$. By theorem 4.3, all derivatives of order greater than n vanish at y , so the analytic Taylor series at y is a polynomial of degree at most n . Therefore up agrees with a polynomial on a neighborhood of y .

Choose inside that neighborhood another reduced dyadic point z of level $m > n$. The same polynomial has zero m -th derivative, whereas (40) says $\text{up}^{(m)}(z) \neq 0$. This contradiction proves the claim. The bridge between F and up transfers the statement to $[0, 1]$. $\square$

Remark 4.5. There is no contradiction with the fact that up is analytic on $(-\infty, -1)$ and $(1, \infty)$: it is identically zero there. The theorem is precisely a statement about every point of the topological support.

5 Fourier analysis, infinite products, and Poisson summation

5.1 The entire Fourier transform

We now collect the Fourier statements used implicitly above. Our convention is

$$\widehat{g}(z) = \int_{\mathbb{R}} g(x) e^{-2\pi i x z} dx. \quad (41)$$

Because $\text{up} \in C_c^\infty(\mathbb{R})$, its transform is entire and rapidly decreasing on every horizontal strip. The refinement equation determines it exactly.

Theorem 5.1 (Sinc product and inversion). *For every $z \in \mathbb{C}$,*

$$\widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \frac{\sin(\pi z/2^j)}{\pi z/2^j}. \quad (42)$$

For every $x \in \mathbb{R}$,

$$\text{up}(x) = \int_{\mathbb{R}} \widehat{\text{up}}(t) e^{2\pi i t x} dt. \quad (43)$$

The product and all of its complex derivatives converge locally uniformly.

Proof. The product identity is (31), with $\widehat{\text{up}}(0) = \int \text{up} = 1$. Local uniform convergence follows from $1 - \text{sinc } z = O(z^2)$ and the convergence of $\sum 4^{-j}$, uniformly on compact sets. Since $\widehat{\text{up}}$ is integrable by theorem 3.1, ordinary Fourier inversion applies. The same decay after multiplication by powers of t justifies differentiation under the inversion integral to every order. $\square$

The first factor already shows that

$$\widehat{\text{up}}(m) = 0 \quad (m \in \mathbb{Z} \setminus \{0\}). \quad (44)$$

The zero multiplicities contain additional arithmetic information. If $m \neq 0$ is an integer and $\nu_2(m)$ is its 2-adic valuation, then exactly the factors with $0 \leq j \leq \nu_2(m)$ vanish, so

$$\text{ord}_{z=m} \widehat{\text{up}}(z) = 1 + \nu_2(m). \quad (45)$$

This leads to a third entire product.

Theorem 5.2 (Three equivalent products). *The following ordered products agree for every $z \in \mathbb{C}$:*

$$\widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^j}\right), \quad (46)$$

$$= \prod_{m=1}^{\infty} \left(\cos \frac{\pi z}{2^m}\right)^m, \quad (47)$$

$$= \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}. \quad (48)$$

The last product is understood in its natural symmetric canonical ordering.

Proof. The elementary identity

$$\text{sinc } z = \cos(z/2) \text{sinc}(z/2) \quad (49)$$

implies, for the first N sinc factors,

$$\prod_{j=0}^{N-1} \text{sinc}\left(\frac{\pi z}{2^j}\right) = \left[\prod_{m=1}^N \left(\cos \frac{\pi z}{2^m}\right)^m \right] \text{sinc}\left(\frac{\pi z}{2^N}\right)^N. \quad (50)$$

The final factor tends to 1 locally uniformly, since $N(1 - \text{sinc}(\pi z/2^N)) = O(N4^{-N})$. This proves (47).

For (48), use the Weierstrass product $\text{sinc}(\pi z) = \prod_{r \geq 1} (1 - z^2/r^2)$ in each sinc factor. A zero at the integer m occurs in the j -th scaled factor precisely when $2^j \mid m$. Thus its total multiplicity is $1 + \nu_2(m)$, as in (45). Regrouping symmetric pairs of zeros gives the stated canonical product. $\square$

5.2 Poisson summation and partition of unity

Since up is a compactly supported smooth function, both it and its transform are Schwartz functions. Poisson summation is therefore available without any conditional convergence issue.

Theorem 5.3 (Poisson summation). *For $a > 0$ and $x \in \mathbb{R}$,*

$$\sum_{m \in \mathbb{Z}} \text{up}(x + am) = \frac{1}{a} \sum_{k \in \mathbb{Z}} \widehat{\text{up}}(k/a) e^{2\pi i k x / a}. \quad (51)$$

Both series converge absolutely and locally uniformly with all derivatives.

Proof. Apply the standard Poisson formula to the Schwartz function $t \mapsto \text{up}(x + at)$. The scaling and translation rules for the Fourier transform give (51). Rapid decay gives the asserted convergence. $\square$

The integer zeros (44) collapse several specializations.

Corollary 5.4 (Integer translates). *For all $x \in \mathbb{R}$,*

$$\sum_{m \in \mathbb{Z}} \text{up}(x + m) = 1. \quad (52)$$

In particular, for $0 \leq x \leq 1$,

$$\text{up}(x) + \text{up}(x - 1) = 1. \quad (53)$$

More generally, for every positive integer n ,

$$\sum_{m \in \mathbb{Z}} \text{up}\left(x + \frac{m}{n}\right) = n. \quad (54)$$

Proof. Set $a = 1$ in (51); only $k = 0$ survives. Support reduces the left side to two terms on $[0, 1]$. For (54), set $a = 1/n$; the dual samples are at kn , again nonzero integers unless $k = 0$. $\square$

The identity (53) is exactly the Fabius symmetry in up coordinates: $\text{up}(x - 1) = F(x)$ and $\text{up}(x) = 1 - F(x)$.

Taking lattice spacing 2 instead gives the rapidly convergent cosine expansion.

Corollary 5.5 (Cosine expansion). *For $|x| \leq 1$,*

$$\text{up}(x) = \frac{1}{2} + \sum_{k=0}^{\infty} \widehat{\text{up}}\left(\frac{2k+1}{2}\right) \cos((2k+1)\pi x). \quad (55)$$

The series and all of its derivatives converge uniformly.

Proof. For $|x| \leq 1$, support makes $\sum_m \text{up}(x + 2m) = \text{up}(x)$. In the dual series, every nonzero even index samples $\widehat{\text{up}}$ at a nonzero integer and vanishes. Pair the terms indexed by $2k + 1$ and $-(2k + 1)$, using that $\widehat{\text{up}}$ is real and even on the real axis. Rapid decay proves smooth uniform convergence. $\square$

5.3 A useful sampling identity

At $x = 0$, (51) reads

$$\sum_{m \in \mathbb{Z}} \text{up}(am) = \frac{1}{a} \sum_{m \in \mathbb{Z}} \widehat{\text{up}}(m/a). \quad (56)$$

When $1/2 \leq a \leq 1$, only the samples $0, \pm a$ of up can be nonzero, so

$$1 + 2 \text{up}(a) = \frac{1}{a} \sum_{m \in \mathbb{Z}} \widehat{\text{up}}(m/a). \quad (57)$$

Equivalently, $a + 2a \text{up}(a) = \sum_m \widehat{\text{up}}(m/a)$. This is one of several ways to turn values of the bump into rapidly convergent transform sums.

6 Moments and generating functions

Two rational moment sequences organize much of the exact arithmetic.

Definition 6.1 (Even moments). For $n \geq 0$, let

$$c_n = \int_{\mathbb{R}} x^{2n} \text{up}(x) \, dx. \quad (58)$$

All odd moments vanish because up is even.

Definition 6.2 (Half-moments). Let $d_0 = 1$, and for $n \geq 1$ put

$$d_n = n \int_0^1 x^{n-1} \text{up}(x) \, dx. \quad (59)$$

6.1 The moment series and the recurrence for c_n

Expanding the Fourier integral at the origin gives

$$\widehat{\text{up}}(z) = \sum_{n=0}^{\infty} (-1)^n \frac{c_n}{(2n)!} (2\pi z)^{2n}. \quad (60)$$

The series is entire because $\widehat{\text{up}}$ is entire.

Theorem 6.3 (Moment recurrence). *The moments are positive rational numbers satisfying*

$$c_0 = 1, \quad (2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k \quad (n \geq 1). \quad (61)$$

Proof. The Fourier refinement equation is

$$\widehat{\text{up}}(z) = \text{sinc}(\pi z) \widehat{\text{up}}(z/2). \quad (62)$$

Insert the power series for $\text{sinc}(\pi z)$ and (60), then compare the coefficient of z^{2n} . One obtains

$$(2n+1)2^{2n}c_n = \sum_{k=0}^n \binom{2n+1}{2k} c_k. \quad (63)$$

The last summand is $(2n+1)c_n$; moving it to the left gives (61). Positivity follows either from the integral definition or inductively from the recurrence, and rationality follows inductively because the coefficients are integers. $\square$

The first values are

$$\begin{aligned} c_0 &= 1, & c_1 &= \frac{1}{9}, & c_2 &= \frac{19}{675}, & c_3 &= \frac{583}{59535}, \\ c_4 &= \frac{132809}{32531625}, & c_5 &= \frac{46840699}{24405225075}, & \dots \end{aligned} \quad (64)$$

6.2 The exponential generating function and the d_n recurrence

Define

$$f(z) = 1 + z \int_0^1 \text{up}(t) e^{zt} \, dt. \quad (65)$$

There are three equivalent interpretations:

$$f(z) = \mathbb{E}(e^{zX}) = e^{z/2} \widehat{\text{up}}\left(\frac{iz}{4\pi}\right) = \sum_{n=0}^{\infty} \frac{d_n}{n!} z^n. \quad (66)$$

The first equality follows by integration by parts and $F(t) = 1 - \text{up}(t)$ on $[0, 1]$; the second follows from $X = (Y + 1)/2$; the third is the definition of the Taylor coefficients and agrees with (59).

Split the random series as $X = U_1/2 + X'/2$, with X' an independent copy of X . Its moment generating function obeys

$$f(2z) = \frac{e^z - 1}{z} f(z), \quad (67)$$

where the quotient is assigned its removable value 1 at $z = 0$.

Theorem 6.4 (Half-moment recurrence). *The half-moments satisfy*

$$d_0 = 1, \quad (n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k \quad (n \geq 1). \quad (68)$$

Proof. Multiply (67) by z : $zf(2z) = (e^z - 1)f(z)$. Substitute the exponential generating series and compare the coefficient of z^{n+1} . The term containing d_n appears on both sides; collecting it gives (68). $\square$

The first values are

$$1, \frac{1}{2}, \frac{5}{18}, \frac{1}{6}, \frac{143}{1350}, \frac{19}{270}, \frac{1153}{23814}, \frac{583}{17010}, \frac{1616353}{65063250}, \dots \quad (69)$$

6.3 Relations between the two moment systems

Theorem 6.5 (Binomial transform). *For every $n \geq 0$,*

$$d_n = 2^{-n} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k. \quad (70)$$

Proof. For $n \geq 1$, integrate (59) by parts:

$$\begin{aligned} d_n &= \int_0^1 \text{up}(t) d(t^n) = - \int_0^1 t^n \text{up}'(t) dt \\ &= 2 \int_0^1 t^n \text{up}(2t-1) dt = \int_{-1}^1 \text{up}(x) \left(\frac{x+1}{2}\right)^n dx. \end{aligned}$$

The boundary term vanishes because $\text{up}(1) = 0$ and $t^n = 0$ at $t = 0$. Expand $(x+1)^n$. Odd moments vanish, while the $2k$ -th moment is c_k . This gives (70). The case $n = 0$ is immediate. $\square$

The inverse-power values of F are precisely normalized half-moments.

Theorem 6.6 (Inverse powers of two). *For every $n \geq 0$,*

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}), \quad F(2^{-n}) = \frac{d_n}{n! 2^{\binom{n}{2}}}. \quad (71)$$

Proof. The case $n = 0$ is $F(1) = d_0 = 1$, so suppose $n \geq 1$. On $[0, 1]$ we have $\text{up}(t) = F(1 - t)$. After the substitution $u = 1 - t$,

$$d_n = n \int_0^1 (1 - u)^{n-1} F(u) du. \quad (72)$$

Set $g(u) = F(u/2^n)$. Repeated differentiation of $F'(x) = 2F(2x)$ gives, for $0 \leq u \leq 1$,

$$g^{(n)}(u) = 2^{-n^2} F^{(n)}(u/2^n) = 2^{-\binom{n}{2}} F(u). \quad (73)$$

All derivatives of F vanish at 0, so $g^{(j)}(0) = 0$ for $0 \leq j < n$. Integrating by parts n times therefore yields

$$\int_0^1 (1 - u)^{n-1} g^{(n)}(u) du = (n - 1)! g(1). \quad (74)$$

Indeed, the boundary terms at $u = 0$ vanish by flatness; those at $u = 1$ vanish until the last integration because of the factor $(1 - u)$, and the last one is $(n - 1)! g(1)$.

Substituting (73) into (72) and using $g(1) = F(2^{-n})$ gives

$$d_n = n 2^{\binom{n}{2}} (n - 1)! F(2^{-n}),$$

which is (71). □

Combining theorems 6.5 and 6.6 gives, for $k \geq 0$,

$$c_k = 2(2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \quad (75)$$

This bridge will reappear in the exact dyadic formulas and in the Legendre coefficients.

6.4 Bernoulli recurrences

The functional equations may also be inverted before coefficients are compared. Using

$$\frac{\pi z}{\sin \pi z} = 1 + \sum_{n \geq 1} (-1)^{n+1} \frac{B_{2n}(2^{2n} - 2)}{(2n)!} (\pi z)^{2n}, \quad (76)$$

$$\frac{z}{e^z - 1} = \sum_{n \geq 0} \frac{B_n}{n!} z^n, \quad (77)$$

coefficient comparison gives the following equivalent recurrences.

Proposition 6.7 (Bernoulli recurrences). *For $n \geq 1$,*

$$c_n = \frac{1}{2^{2n} - 1} \sum_{k=1}^n 2^{2n-2k} (2^{2k} - 2) \binom{2n}{2k} B_{2k} c_{n-k}, \quad (78)$$

and

$$d_n = \frac{n 2^{n-2}}{2^n - 1} d_{n-1} - \frac{1}{2^n - 1} \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{2k} 2^{n-2k} B_{2k} d_{n-2k}. \quad (79)$$

Proof. Rewrite (62) as $(\pi z / \sin \pi z) \widehat{\text{up}}(z) = \widehat{\text{up}}(z/2)$ and (67) as $(z / (e^z - 1)) f(2z) = f(z)$. Insert (76)–(77), then compare coefficients. The odd Bernoulli numbers beyond B_1 vanish, producing the displayed forms. □

7 Exact evaluation at dyadic rationals

Dyadic points are exceptional in two complementary ways. Analytically, every sufficiently high derivative vanishes there. Arithmetically, the value is rational and can be computed by finite exact operations. We now prove two formulas: a closed sum that displays the Thue–Morse and moment structure, and a recursive formula adapted to computation.

7.1 The closed Thue–Morse–moment formula

Theorem 7.1 (Closed dyadic formula). *Let $n, a \in \mathbb{N}$. Then the signed global extension satisfies*

$$\Theta\left(\frac{a}{2^n}\right) = \frac{2^{-(\binom{n+1}{2})}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} (2a - 2h - 1)^{n-2k} c_k. \quad (80)$$

The outer sum is empty when $a = 0$. If $0 \leq a \leq 2^n$, then $\Theta(a/2^n) = F(a/2^n)$, so (80) is an exact formula for the bounded Fabius function. If $-2^n \leq q \leq 2^n$, the up-value $\text{up}(q/2^n)$ is obtained by taking $a = q + 2^n$.

Proof. All derivatives of Θ vanish at 0. Taylor's theorem with integral remainder, applied through order n , therefore gives

$$\Theta(x) = \frac{1}{n!} \int_0^x (x-t)^n \Theta^{(n+1)}(t) dt. \quad (81)$$

Put $x = a/2^n$ and use (35):

$$\Theta^{(n+1)}(t) = 2^{\binom{n+2}{2}} \Theta(2^{n+1}t).$$

With $u = 2^{n+1}t$, equation (81) becomes

$$\Theta\left(\frac{a}{2^n}\right) = \frac{2^{-(\binom{n+1}{2})}}{n!} \int_0^{2a} (2a-u)^n \Theta(u) du. \quad (82)$$

Split $[0, 2a]$ into the bands $[2h, 2h+2]$. By (10), and with $v = u - 2h - 1$,

$$\begin{aligned} \int_{2h}^{2h+2} (2a-u)^n \Theta(u) du &= \varepsilon_h \int_{-1}^1 (2a-2h-1-v)^n \text{up}(v) dv \\ &= \varepsilon_h \sum_{j=0}^n \binom{n}{j} (2a-2h-1)^{n-j} (-1)^j \int_{-1}^1 v^j \text{up}(v) dv. \end{aligned}$$

Odd moments vanish, and the $2k$ -th moment is c_k . The surviving signs are $(-1)^{2k} = 1$, producing precisely (80). $\square$

Corollary 7.2 (Rationality). *The values $F(a/2^n)$ and $\text{up}(q/2^n)$ are rational at every dyadic point in their respective domains. The signed evaluator $\Theta(a/2^n)$ is rational for every nonnegative integer numerator a .*

Proof. The moments c_k are rational by theorem 6.3; every other quantity in (80) is integral or a power of 2 divided by a factorial. $\square$

The formula is closed but not necessarily the fastest computational route. It is, however, ideal for proving denominator bounds because the only nonintegral inputs are the finitely many c_k with $k \leq \lfloor n/2 \rfloor$.

7.2 A complete level-four table

At denominator 16, every value has common denominator 2 073 600:

$$F(a/16) = \frac{N_a}{2\,073\,600}. \quad (83)$$

The exact numerators are

a	0	1	2	3	4	5	6	7	8
N_a	0	143	7 200	46 657	144 000	305 857	525 600	777 743	1 036 800

a	9	10	11	12	13	14	15	16
N_a	1 295 857	1 548 000	1 767 743	1 929 600	2 026 943	2 066 400	2 073 457	2 073 600

The symmetry $N_{16-a} = 2\,073\,600 - N_a$ is just $F(1-x) = 1 - F(x)$.

7.3 The inverse-power lookup table

The first values furnished by (71) are

n	d_n	$F(2^{-n})$
0	1	1
1	1/2	1/2
2	5/18	5/72
3	1/6	1/288
4	143/1350	143/2 073 600
5	19/270	19/33 177 600
6	1153/23814	1153/561 842 749 440
7	583/17010	583/179 789 679 820 800

The table can be generated in $O(N^2)$ rational operations from (68).

7.4 Taylor reduction by the highest binary digit

The next formula was designed for exact evaluation. It removes the largest power of two present in the argument and replaces the function value by one at a smaller argument plus a finite polynomial whose coefficients come from the lookup table.

Theorem 7.3 (Taylor-block reduction). *Let $x > 0$, and let $n \in \mathbb{Z}$ be the unique integer such that*

$$2^{-n} \leq x < 2^{-n+1}. \quad (84)$$

Put $y = x - 2^{-n}$. Then

$$\Theta(x) = -\Theta(y) + \begin{cases} \sum_{k=0}^n 2^{\binom{k+1}{2} - \binom{n-k}{2}} \frac{d_{n-k}}{(n-k)!} \frac{y^k}{k!}, & n \geq 0, \\ 0, & n < 0. \end{cases} \quad (85)$$

For $0 < x \leq 1$, one always has $n \geq 0$ and $\Theta(x) = F(x)$.

Proof. Assume first that $n \geq 0$, set $a = 2^{-n}$, and apply Taylor's theorem at a to order n :

$$\Theta(a + y) = \sum_{k=0}^n \frac{\Theta^{(k)}(a)}{k!} y^k + \frac{1}{n!} \int_a^{a+y} (a + y - t)^n \Theta^{(n+1)}(t) dt. \quad (86)$$

The derivatives at the base point are exact inverse-power values:

$$\begin{aligned} \Theta^{(k)}(2^{-n}) &= 2^{\binom{k+1}{2}} \Theta(2^{k-n}) \\ &= 2^{\binom{k+1}{2}} F(2^{-(n-k)}) \\ &= 2^{\binom{k+1}{2} - \binom{n-k}{2}} \frac{d_{n-k}}{(n-k)!}, \quad 0 \leq k \leq n. \end{aligned} \quad (87)$$

It remains to identify the integral remainder. Write $t = a + s$. Since $0 \leq s \leq y < a$, we have $0 \leq 2^{n+1}s < 2$. Using the derivative dilation and the band sign $\Theta(2 + z) = -\Theta(z)$ for $0 \leq z \leq 2$,

$$\begin{aligned} \Theta^{(n+1)}(a + s) &= 2^{\binom{n+2}{2}} \Theta(2^{n+1}(a + s)) \\ &= 2^{\binom{n+2}{2}} \Theta(2 + 2^{n+1}s) \\ &= -2^{\binom{n+2}{2}} \Theta(2^{n+1}s) = -\Theta^{(n+1)}(s). \end{aligned}$$

Therefore the remainder in (86) equals

$$-\frac{1}{n!} \int_0^y (y - s)^n \Theta^{(n+1)}(s) ds. \quad (88)$$

Taylor's theorem at 0, where all derivatives vanish, identifies (88) with $-\Theta(y)$. Substitution of (87) proves the first case.

If $n < 0$, then 2^{-n} is an even positive integer. Formula (35) shows that every derivative of Θ at that point is zero. Taylor's theorem with any order, together with the same translated-remainder argument, gives $\Theta(x) = -\Theta(y)$, which is the second case. $\square$

Remark 7.4 (Why the recursion is binary). If $x = a/2^N \in (0, 1]$ and 2^b is the highest power of two not exceeding a , then $2^{-n} = 2^b/2^N$ with $n = N - b$, and

$$y = \frac{a - 2^b}{2^N}.$$

Thus one application of (85) clears the highest 1-bit of a . Repetition terminates after exactly $w_2(a)$ steps.

Algorithm 7.5 (Exact dyadic evaluator). To compute $F(a/2^N)$ exactly for integers $0 \leq a \leq 2^N$:

1. Precompute $d_0, \dots, d_N$ from (68), or equivalently precompute $F(2^{-m})$ from (71).
2. If $a = 0$, return 0; if $a = 2^N$, return 1.
3. Let $b = \lfloor \log_2 a \rfloor$, $n = N - b$, and $a' = a - 2^b$. Set $y = a'/2^N$.
4. Evaluate the degree- n polynomial in (85), preferably by Horner's rule, and subtract the recursively computed value $F(y)$.

The precomputation takes $O(N^2)$ rational operations. The recursive evaluation takes $O(Nw_2(a))$ further operations in the simplest implementation; more careful reuse of factorials and powers reduces constants substantially.

Example 7.6 (The exact value at $5/16$). The highest power of two in the numerator 5 is 4, so $x = 5/16 = 1/4 + 1/16$. Here $n = 2$ and $y = 1/16$. The polynomial in (85) is

$$\begin{aligned} 2^{-1} \frac{d_2}{2!} + 2^1 \frac{d_1}{1!} y + 2^3 \frac{d_0}{0!} \frac{y^2}{2!} \\ = \frac{5}{72} + \frac{1}{16} + \frac{1}{64}. \end{aligned}$$

Since $F(1/16) = 143/2\,073\,600$,

$$F(5/16) = -\frac{143}{2\,073\,600} + \frac{5}{72} + \frac{1}{16} + \frac{1}{64} = \frac{305\,857}{2\,073\,600}. \quad (89)$$

This agrees with the level-four table above.

7.5 Correctness under rescaling and clamping

An implementation must not depend on whether the input fraction was reduced. The analytic identity is trivial,

$$F\left(\frac{a}{2^N}\right) = F\left(\frac{2a}{2^{N+1}}\right), \quad (90)$$

but an exact evaluator should satisfy it by its own arithmetic recursion. The formal development proves precisely this prefix/refinement invariance, including all boundary cases. It also proves the clamping laws

$$F(a/2^N) = 0 \quad (a \leq 0), \quad F(a/2^N) = 1 \quad (a \geq 2^N) \quad (91)$$

for the bounded evaluator, while the signed evaluator continues according to (10). These statements ensure that unreduced representations, leading binary zeroes, and out-of-range inputs cannot change the mathematical answer.

7.6 Exact dyadic derivatives

The value formula extends immediately to all derivatives. For $0 \leq x \leq 1$ and $k \geq 0$,

$$F^{(k)}(x) = 2^{\binom{k+1}{2}} \Theta(2^k x). \quad (92)$$

At $x = a/2^N$ the right side is another exact signed dyadic value, now at level $N - k$. If a is odd, the derivative of order N is nonzero and every higher derivative is zero. Thus the exact evaluator computes not merely values but the entire finite Taylor jet at every reduced dyadic point.

8 Arithmetic of the moments and dyadic values

The rational recurrences hide unexpectedly integral sequences. Throughout this section, empty products are 1, and $(-1)!! = 1$.

8.1 Integral moment numerators

Definition 8.1. Define integers F_n and G_n implicitly by

$$c_n = \frac{F_n}{(2n+1)!! \prod_{v=1}^n (2^{2v} - 1)}, \quad (93)$$

$$d_n = \frac{G_n}{(n+1)! \prod_{k=1}^n (2^k - 1)}. \quad (94)$$

Theorem 8.2 (Integrality of F_n and G_n). *The numbers F_n and G_n are positive integers. They satisfy the division-free recurrences*

$$F_n = \sum_{k=0}^{n-1} F_k \binom{2n+1}{2k} \frac{(2n-1)!!}{(2k+1)!!} \prod_{v=k+1}^{n-1} (2^{2v} - 1) \quad (n \geq 1), \quad (95)$$

and

$$G_n = \sum_{k=0}^{n-1} G_k \binom{n+1}{k} \frac{n!}{(k+1)!} \prod_{j=k+1}^{n-1} (2^j - 1) \quad (n \geq 1). \quad (96)$$

Proof. Substitute (93) into the recurrence (61). After cancellation one obtains (95); every quotient of odd double factorials is an integer, so induction proves $F_n \in \mathbb{N}$. Positivity is immediate. The same substitution of (94) into (68) gives (96); the factorial quotient and product are integral, so induction proves the assertion for G_n . $\square$

The first terms are

n	F_n	G_n
0	1	1
1	1	1
2	19	5
3	2 915	84
4	2 788 989	4 004
5	14 754 820 185	494 760
6	402 830 065 455 939	150 120 600
7	54 259 734 183 964 303 995	107 969 547 840
8	34 931 036 957 548 128 175 343 565	179 605 731 622 464

Equation (71) and (94) give the exact closed form

$$F(2^{-n}) = \frac{2^{-\binom{n}{2}} G_n}{n!(n+1)! \prod_{k=1}^n (2^k - 1)}. \quad (97)$$

For odd indices, the two integral sequences are related by

$$\frac{G_{2n+1}}{2n+1} = 2^n (n+1)! F_n \prod_{k=0}^n (2^{2k+1} - 1). \quad (98)$$

8.2 Parity of the moment numerators

Theorem 8.3 (Oddness of F_n). *Every F_n is odd.*

Proof. Assume inductively that $F_0, \dots, F_{n-1}$ are odd. Every factor in (95) other than the binomial coefficient is odd. Hence the parity of F_n is the parity of the number of odd coefficients $\binom{2n+1}{2k}$ with $0 \leq k < n$.

Write n in binary. Lucas's theorem says that $\binom{2n+1}{2k}$ is odd precisely when every 1-bit of k occurs in a position where n also has a 1-bit. There are $2^{w_2(n)}$ such k in the range $0 \leq k \leq n$. The omitted coefficient at $k = n$ is odd, leaving $2^{w_2(n)} - 1$ odd summands. Since $n \geq 1$ implies $w_2(n) \geq 1$, this number is odd. Thus F_n is odd. The initial value $F_0 = 1$ starts the induction. $\square$

8.3 Reshetnikov's numbers

Definition 8.4 (Reshetnikov numbers). For $n \geq 1$, define

$$R_n = 2^{\binom{n-1}{2}} (2n)! F(2^{-n}) \prod_{m=1}^{\lfloor n/2 \rfloor} (2^{2m} - 1). \quad (99)$$

Substituting (71) gives the simpler identity

$$R_n = 2d_n (2n-1)!! \prod_{m=1}^{\lfloor n/2 \rfloor} (2^{2m} - 1). \quad (100)$$

All factors after $2d_n$ are odd.

Theorem 8.5 (Integrality of the Reshetnikov numbers). *For every $n \geq 1$, R_n is a positive integer. More explicitly,*

$$R_{2n+1} = F_n \frac{(4n+1)!!}{(2n-1)!!}, \quad n \geq 0, \quad (101)$$

and, for $n \geq 1$,

$$R_{2n} = \sum_{k=0}^n \frac{2F_k}{2^{2n}} \binom{2n}{2k} \frac{(4n-1)!!}{(2k+1)!!} \prod_{\ell=k+1}^n (2^{2\ell} - 1). \quad (102)$$

Although the individual terms in (102) need not be integers, the sum is.

Proof. Evenness of up gives $d_{2n+1} = (2n+1)c_n/2$. Substitute this and (93) into (100); cancellation gives (101), proving the odd-indexed case.

For the even formula, insert the binomial transform (70) into (100), then replace each c_k by (93). This yields (102); all possible nonintegral denominators are powers of 2.

It therefore suffices to prove that the 2-adic valuation of R_{2n} is nonnegative. Since the extra factor in (100) is odd, $v_2(R_m) = v_2(2d_m)$. The odd-indexed case has already shown nonnegative valuation there. For an even index, multiply (68) by 2 and write it at $2n$:

$$(2n+1)(2^{2n}-1)2d_{2n} = \sum_{k=0}^{2n-1} \binom{2n+1}{k} 2d_k. \quad (103)$$

The left prefactor is odd. By induction every term on the right is a 2-adically integral rational number, so their sum is too. Hence $v_2(2d_{2n}) \geq 0$, proving integrality. Positivity follows from the defining formula. $\square$

The first terms are

$$1, 5, 15, 1\,001, 5\,985, 2\,853\,675, 26\,261\,235, 72\,808\,620\,885, 915\,304\,354\,965, \dots \quad (104)$$

The integrality theorem can be sharpened to exact parity.

Theorem 8.6 (Exact 2-adic valuation). *For every $n \geq 1$,*

$$v_2(2d_n) = 0, \quad (105)$$

so R_n is odd. Equivalently,

$$v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!). \quad (106)$$

Proof. For odd indices, (101) and theorem 8.3 show that R_{2m+1} is odd; the odd factor in (100) then gives (105).

Proceed by induction for even indices. Let

$$N_m = (2m - 1)!! \prod_{j=1}^{\lfloor m/2 \rfloor} (2^{2j} - 1).$$

The divisibility $N_k \mid N_{2n}$ for $k \leq 2n$ and the integrality of the R_k show that every $2d_k N_{2n}$ is an integer. By the induction hypothesis it is odd for $1 \leq k < 2n$, while the $k = 0$ term is even. Multiply (103) by N_{2n} . Lucas's theorem shows that an odd number of coefficients $\binom{2n+1}{k}$ with $1 \leq k < 2n$ are odd. Consequently the right-hand side is odd, and so $2d_{2n} N_{2n}$ is odd. As N_{2n} is odd, $v_2(2d_{2n}) = 0$.

Finally, apply v_2 to $F(2^{-n}) = d_n / (n! 2^{\binom{n}{2}})$. Equation (105) says $v_2(d_n) = -1$, giving (106). $\square$

8.4 Common denominators on a dyadic grid

For $n \geq 1$, let D_n be the least common multiple of the denominators, in lowest terms, of

$$F\left(\frac{2k+1}{2^n}\right), \quad 0 \leq k < 2^{n-1}, \quad (107)$$

and set $D_0 = 1$. By symmetry, the same number is obtained from the corresponding positive odd-grid values of up.

Theorem 8.7 (Uniform denominator bound). *Let $m = \lfloor n/2 \rfloor$. Then*

$$D_n \mid n! 2^{\binom{n+1}{2}} (2m+1)!! \prod_{k=1}^m (2^{2k} - 1). \quad (108)$$

More generally, for every integer a with $|a| < 2^n$,

$$\text{up}(a/2^n) n! 2^{\binom{n+1}{2}} (2m+1)!! \prod_{k=1}^m (2^{2k} - 1) \in \mathbb{N} \cup \{0\}. \quad (109)$$

Proof. The closed dyadic formula shows that after multiplication by $n! 2^{\binom{n+1}{2}}$, every level- n value is an integer linear combination of $c_0, \dots, c_m$. By (93), for each $k \leq m$,

$$c_k (2m+1)!! \prod_{r=1}^m (2^{2r} - 1)$$

is an integer: the denominator in (93) divides the displayed common factor. This proves (109) and hence the LCM bound (108). $\square$

The first common denominators are

$$\begin{aligned} D_0, \dots, D_9 = & 1, 2, 72, 288, 2\,073\,600, 33\,177\,600, \\ & 2\,809\,213\,747\,200, 179\,789\,679\,820\,800, \\ & 704\,200\,217\,922\,109\,440\,000, \\ & 180\,275\,255\,788\,060\,016\,640\,000. \end{aligned} \quad (110)$$

The denominator of $F(2^{-n})$ always divides D_n , and theorem 8.6 determines its exact power of 2.

8.5 The remaining denominator conjecture

The following experimentally observed structure is included in the formal repository as a conjecture, not as a theorem.

Conjecture 8.8 (Arias de Reyna–Reshetnikov denominator pattern). *Set*

$$A_n = 2^{-\binom{n}{2}} D_n. \quad (111)$$

Then:

1. $A_{2n} = A_{2n+1}$ for $n \geq 1$.
2. If $K_n = A_{2n-1}$, then $2(2n-1)! \mid K_n$.
3. The integers

$$H_n = \frac{K_n}{2(2n-1)!} \quad (112)$$

are odd.

Equivalently, if true,

$$D_{2n-1} = 2^{1+\binom{2n-1}{2}} (2n-1)! H_n, \quad D_{2n} = 2^{1+\binom{2n}{2}} (2n+1)! H_{n+1}. \quad (113)$$

The first H_n would be

$$1, 3, 135, 8\,505, 3\,614\,625, 2\,218\,656\,825, \dots$$

The proven uniform bound (108) and the exact 2-adic valuation (106) constrain the conjecture but do not settle its odd prime divisibility.

9 Finite q -binomial and Thue–Morse formulas

The moment formula (80) is not the only finite expression. There is a second family in which a Gaussian binomial transform isolates the desired coefficient of the refinement series.

For $q \in \mathbb{Q}$, write

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}. \quad (114)$$

We use $q = 1/2$ below.

Theorem 9.1 (q -binomial formula at an inverse power). *For every $n \geq 0$,*

$$\begin{aligned} F(2^{-n}) &= \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{\left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ &\quad \times \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k)^{n+k}. \end{aligned} \quad (115)$$

The centered power $r - 2^k$ may be replaced simultaneously in every inner sum by $r - 2^k + c$ for any rational constant c . In particular, the frequently used half-shift $c = 1/2$ gives the same value.

Proof. We give the coefficient-extraction argument. For fixed k , the exponential generating function of the centered Thue–Morse power sums is

$$\begin{aligned} \sum_{d \geq 0} \frac{z^d}{d!} \sum_{r=0}^{2^k-1} \varepsilon_r (r - 2^k)^d &= e^{-2^k z} \sum_{r=0}^{2^k-1} \varepsilon_r e^{rz} \\ &= e^{-2^k z} \prod_{j=0}^{k-1} (1 - e^{2^j z}). \end{aligned} \quad (116)$$

The last equality follows by expanding the product: choosing the exponential term in the j -th factor records the j -th binary digit and contributes a minus sign.

Normalize the half-moment series by

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad a_n = \frac{d_n}{n!}. \quad (117)$$

The functional equation (67) is a formal power-series refinement equation for A . Iterating it k times expresses a rescaled copy of A as A multiplied by exactly the product of exponential-difference factors in (116). Thus the coefficient appearing in the k -th inner sum can be written as the coefficient of a common recurrence series multiplied by a k -dependent refinement factor.

The finite q -binomial identity

$$\sum_{k=0}^n (-1)^k q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q x^k = \prod_{j=0}^{n-1} (1 - xq^j) \quad (118)$$

annihilates every node power of degree less than n when evaluated at the dyadic nodes arising from the refinement factors. At degree n it leaves $2^{\binom{n+1}{2}} (1/2; 1/2)_n$. Consequently the weighted outer sum in (115) isolates the n -th coefficient a_n , with the displayed normalization. Since $F(2^{-n}) = 2^{-\binom{n}{2}} d_n/n!$, simplification gives (115).

For translation invariance, replacing every inner power by $(r - 2^k + c)^d$ multiplies the generating function (116) by e^{cz} . Coefficient convolution introduces only lower-degree weighted sums; all of those are killed by (118). The isolated degree- n coefficient is therefore unchanged. $\square$

Theorem 9.2 (Arbitrary numerator). *For $m, n \in \mathbb{N}$ and $c \in \mathbb{Q}$, define*

$$\begin{aligned} Q_{n,m,c} &= \frac{1}{2^{n^2} (1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}} (n+k)!} \\ &\quad \times \sum_{r=0}^{m2^k-1} \varepsilon_r (r - m2^k + c)^{n+k}. \end{aligned} \quad (119)$$

Then $Q_{n,m,c}$ is independent of c and

$$Q_{n,m,c} = \Theta(m/2^n). \quad (120)$$

In particular, if $0 \leq m \leq 2^n$, it equals $F(m/2^n)$.

Proof. The exponential identity (116) has a block version in which the range $0 \leq r < m2^k$ is partitioned into m consecutive Thue–Morse blocks. The same formal refinement calculation now produces the signed global translate at $m/2^n$. The q -binomial annihilation is unchanged, because it depends only on the scale index k and the target coefficient n . A common translation again multiplies every block series by e^{cz} and therefore contributes only annihilated lower coefficients. This proves both claims. $\square$

Remark 9.3. The formula is finite but its inner ranges grow like $m2^k$. For a single dyadic value, the highest-bit Taylor recursion is usually computationally superior. The q -binomial form is valuable because it reveals a different algebraic mechanism: Thue–Morse cancellation supplies high-order finite differences, while Gaussian binomial weights select one coefficient of the refinement equation.

10 Legendre expansion and best polynomial approximation

Because up is even, its Legendre expansion contains only even degrees. Let P_j denote the ordinary Legendre polynomial normalized by $P_j(1) = 1$.

Definition 10.1 (Legendre coefficients). For $n \geq 0$, set

$$u_n = \frac{4n+1}{2} \int_{-1}^1 \text{up}(x) P_{2n}(x) dx. \quad (121)$$

The coefficient of P_{2n+1} is zero by parity. The even coefficients admit an exact formula involving only inverse-power dyadic values.

Theorem 10.2 (Exact Legendre coefficient). For every $n \geq 0$,

$$u_n = 4^{-n} (4n+1) \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} (2k)! \times 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \quad (122)$$

Proof. Use the even-power expansion

$$P_{2n}(x) = 4^{-n} \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} x^{2k}. \quad (123)$$

Insert it into (121) and integrate term by term. The integral of $x^{2k} \text{up}(x)$ is c_k . Finally substitute (75); the factor 2 there cancels the normalization $1/2$ in (121), giving (122). $\square$

Thus every Legendre coefficient is rational, and the exact dyadic evaluator gives a purely rational algorithm for computing it.

Theorem 10.3 (Absolute and uniform Legendre expansion). On $[-1, 1]$,

$$\text{up}(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x), \quad (124)$$

with absolute and uniform convergence. The series may be differentiated termwise any fixed number of times after taking sufficiently many integrations by parts; in particular it converges correctly at $x = \pm 1$.

Proof. Let

$$\mathcal{L}g = -\frac{d}{dx} \left((1-x^2)g'(x) \right). \quad (125)$$

Then $\mathcal{L}P_j = j(j+1)P_j$, and $\mathcal{L}$ is formally self-adjoint on $[-1, 1]$. Boundary terms vanish because of the factor $1-x^2$; the flatness of up makes every repeated integration legitimate. Hence, for every M ,

$$\int_{-1}^1 \text{up}(x) P_j(x) dx = \frac{1}{[j(j+1)]^M} \int_{-1}^1 (\mathcal{L}^M \text{up})(x) P_j(x) dx. \quad (126)$$

Since $|P_j(x)| \leq 1$ on $[-1, 1]$, the coefficient is $O_M(j^{1-2M})$ after including the standard normalization $(2j+1)/2$. Thus the coefficients decay faster than every power. The Weierstrass test gives absolute uniform convergence. Completeness of the Legendre basis in $L^2[-1, 1]$ identifies the uniform limit with up . $\square$

Theorem 10.4 (Least-squares optimality). *Let*

$$S_N(x) = \sum_{n=0}^N u_n P_{2n}(x). \quad (127)$$

Among all real polynomials of degree at most $2N+1$, S_N is the unique minimizer of

$$\int_{-1}^1 |\text{up}(x) - p(x)|^2 dx. \quad (128)$$

Proof. The Legendre polynomials form an orthogonal basis. Any polynomial of degree at most $2N+1$ decomposes uniquely into its projection onto $\text{span}\{P_0, \dots, P_{2N+1}\}$; the orthogonal projection of up is its truncated Legendre series. Odd coefficients vanish, so that projection is exactly S_N . Pythagoras gives

$$\|\text{up} - p\|_2^2 = \|\text{up} - S_N\|_2^2 + \|S_N - p\|_2^2,$$

which proves uniqueness and optimality. $\square$

On $[0, 2]$, the signed extension satisfies $\Theta(x) = \text{up}(x-1)$. Therefore

$$\Theta(x) = \sum_{n \geq 0} u_n P_{2n}(x-1), \quad 0 \leq x \leq 2. \quad (129)$$

If ordinary powers are desired, one may use

$$P_{2n}(x-1) = \sum_{j=0}^{2n} (-1)^j 2^{-j} \binom{2n}{j} \binom{2n+j}{j} x^j. \quad (130)$$

This converts every finite Legendre approximant into an exact rational polynomial in x .

11 Sharp asymptotics at the flat endpoint

The value $F(x)$ tends to zero faster than every power as $x \downarrow 0$, because all derivatives at 0 vanish. The actual scale is much subtler than a simple stretched exponential. A logarithmic saddle interacts with the dyadic refinement to produce a small but genuine periodic fluctuation.

Throughout this section, set

$$L = \log 2. \quad (131)$$

We write γ for Euler's constant and use the Stieltjes convention

$$\zeta(1+s) = \frac{1}{s} + \sum_{n=0}^{\infty} \frac{(-1)^n \gamma_n}{n!} s^n, \quad \gamma_0 = \gamma. \quad (132)$$

Thus $\gamma_1 \approx -0.07281584548$.

11.1 The negative Laplace product

Let $G(z) = f(z) = \mathbb{E}(e^{zX})$ be the entire moment generating function from (66). For $s > 0$, independence in (11) gives

$$G(-s) = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}. \quad (133)$$

Every factor lies in $(0, 1)$, so define the real logarithm

$$q(s) = \log G(-s) = \sum_{j=1}^{\infty} \log \left(\frac{1 - e^{-s/2^j}}{s/2^j} \right). \quad (134)$$

The product refinement becomes the exact dilation law

$$q(2s) = q(s) + \log \left(\frac{1 - e^{-s}}{s} \right). \quad (135)$$

The exponentially small forward tail is

$$T(s) = \sum_{m=0}^{\infty} \log(1 - e^{-s2^m}), \quad E(s) = -T(s) > 0. \quad (136)$$

It satisfies

$$T(2s) = T(s) - \log(1 - e^{-s}). \quad (137)$$

Lemma 11.1 (Forward-tail bound). *For every $s > 0$,*

$$0 < E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \quad (138)$$

In particular, for $s \geq \log 2$,

$$E(s) \leq 4e^{-s}. \quad (139)$$

Proof. For $0 < u < 1$, $-\log(1 - u) \leq u/(1 - u)$. Put $y = e^{-s}$. Since $2^m \geq m + 1$,

$$E(s) \leq \sum_{m=0}^{\infty} \frac{y^{2^m}}{1 - y^{2^m}} \leq \frac{1}{1 - y} \sum_{m=0}^{\infty} y^{m+1} = \frac{y}{(1 - y)^2}.$$

If $y \leq 1/2$, the final denominator is at least $1/4$. □

11.2 The exact periodic decomposition

Write $t = \log_2 s = (\log s)/L$ and define

$$R(t) = q(2^t) + \frac{L}{2}(t^2 - t) + T(2^t). \quad (140)$$

Theorem 11.2 (Exact log-periodic decomposition). *The function R is smooth and one-periodic. For every $s > 0$,*

$$q(s) = -\frac{(\log s)^2}{2L} + \frac{1}{2} \log s + R(\log_2 s) + E(s). \quad (141)$$

Proof. Use (135), (137), and $\log s = Lt$:

$$\begin{aligned} R(t+1) - R(t) &= \log\left(\frac{1-e^{-s}}{s}\right) + \frac{L}{2}((t+1)^2 - (t+1) - t^2 + t) \\ &\quad - \log(1-e^{-s}) \\ &= -\log s + Lt = 0. \end{aligned}$$

Solving (140) for $q(s)$ gives (141). Smoothness follows by locally uniform termwise differentiation of the defining series; the exponentially decaying tail causes no difficulty. $\square$

Let

$$C_{\Gamma\zeta} = \frac{\gamma^2}{2} + \gamma_1 - \frac{\pi^2}{12} \approx -0.728693917003930606. \quad (142)$$

Theorem 11.3 (Mean and Fourier coefficients). *The mean of R is*

$$\bar{R} := \int_0^1 R(t) dt = \frac{C_{\Gamma\zeta}}{L} - \frac{L}{12}. \quad (143)$$

Define the centered fluctuation

$$\Psi(t) = R(t) - \bar{R}. \quad (144)$$

Then Ψ is smooth, one-periodic, has mean zero, and for $k \neq 0$ its Fourier coefficient is

$$\hat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}, \quad \chi_k = \frac{2\pi ik}{L}. \quad (145)$$

Consequently

$$\Psi(t) = 2\Re \sum_{k=1}^{\infty} \left[-\frac{\Gamma(-2\pi ik/L)\zeta(1-2\pi ik/L)}{L} \right] e^{2\pi ikt}, \quad (146)$$

with absolute convergence after any number of termwise derivatives. The function Ψ is not constant.

Proof. The Mellin kernel behind (134) is $-\log(1-e^{-x})$. For $\Re z > 0$,

$$\int_0^{\infty} x^{z-1} [-\log(1-e^{-x})] dx = \Gamma(z)\zeta(z+1), \quad (147)$$

obtained by expanding $-\log(1-e^{-x}) = \sum_{m \geq 1} e^{-mx}/m$ and integrating termwise. Periodizing on the multiplicative lattice $2^{\mathbb{Z}}$ turns Mellin frequencies into ordinary Fourier frequencies $\chi_k = 2\pi ik/L$. For $k \neq 0$, evaluation of the Mellin transform at $-\chi_k$, followed by the elementary integration by parts that converts $\zeta(1-\chi_k)$ to the logarithmic kernel, gives (145).

At frequency zero, one takes the finite part of $\Gamma(z)\zeta(1+z)$. The Laurent expansions

$$\begin{aligned} \Gamma(z) &= \frac{1}{z} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12} \right) z + O(z^2), \\ \zeta(1+z) &= \frac{1}{z} + \gamma - \gamma_1 z + O(z^2) \end{aligned}$$

show that the constant term of their product is $-C_{\Gamma\zeta}$. Combining this finite part with the mean of the quadratic normalization in (140) yields (143).

Stirling's formula gives exponential decay of $\Gamma(-\chi_k)$ in $|k|$, while $\zeta(1-\chi_k)$ has at most subexponential growth. Hence the Fourier series is smoothly absolutely convergent. The coefficient at $k = 1$ is nonzero: the Gamma function has no zeros, and the zeta function has no zeros on $\Re s = 1$ away from its pole. Thus Ψ is nonconstant. $\square$

Numerically,

$$\widehat{\Psi}(1) \approx (7.55865093976410 - 7.47009091179162i) \times 10^{-7}, \quad (148)$$

while $|\widehat{\Psi}(2)| \approx 6.69 \times 10^{-13}$. The oscillation is tiny, but it is not asymptotically negligible relative to an error tending to zero.

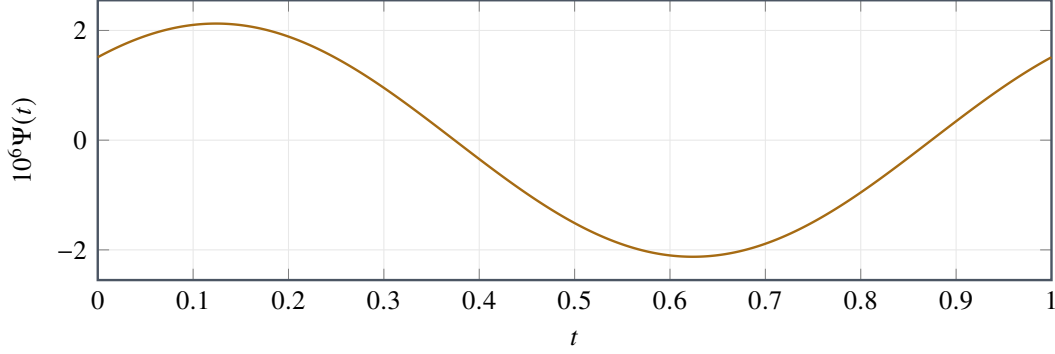


Figure 3: The centered periodic correction, magnified by 10^6 . The first Fourier harmonic already gives visual accuracy.

11.3 The lower Lambert phase

The right coordinate is the large solution of

$$\lambda 2^{-\lambda} = x. \quad (149)$$

For $0 < Lx < e^{-1}$, define

$$\lambda(x) = -\frac{W_{-1}(-Lx)}{L}, \quad r(x) = 2^{\lambda(x)}. \quad (150)$$

Here W_{-1} is the lower real branch of Lambert's W function.

Proposition 11.4 (Exact saddle coordinates). *For sufficiently small $x > 0$,*

$$\lambda(x) 2^{-\lambda(x)} = x, \quad r(x)x = \lambda(x), \quad \log r(x) = L\lambda(x). \quad (151)$$

Moreover $\lambda(x) \rightarrow \infty$ as $x \downarrow 0$ and

$$\lambda(x) \sim \frac{-\log x}{L}. \quad (152)$$

Proof. The defining identity $W(z)e^{W(z)} = z$ with $W = W_{-1}(-Lx)$ gives $(-L\lambda)e^{-L\lambda} = -Lx$, proving the first formula. The other two are immediate. The lower branch tends to $-\infty$ as its argument approaches 0 from below, so $\lambda \rightarrow \infty$. Taking logarithms in (149) gives

$$L\lambda - \log \lambda = -\log x, \quad (153)$$

from which (152) follows. $\square$

For explicit conversion to elementary logarithms, put

$$A = -\log(Lx), \quad B = \log A. \quad (154)$$

The solution $y = L\lambda$ of $y - \log y = A$ has the standard expansion

$$L\lambda = A + B + \frac{B}{A} + \frac{B - \frac{1}{2}B^2}{A^2} + \frac{B - \frac{3}{2}B^2 + \frac{1}{3}B^3}{A^3} + O\left(\frac{B^4}{A^4}\right). \quad (155)$$

The formal development proves the corresponding expansion to arbitrary order, with explicit recursively generated polynomials in B .

For a dyadic argument $x = 2^{-t}$, the phase $\lambda_t = \lambda(2^{-t})$ satisfies the exact fixed-point relation

$$\lambda_t - \frac{\log \lambda_t}{L} = t, \quad (156)$$

and therefore

$$\lambda_t = t + \frac{\log t}{L} + o(1). \quad (157)$$

The fractional part relevant to the periodic correction is $\{\lambda_t\}$, not $\{t\}$.

11.4 Exact Bromwich reduction

The Laplace transform of the clamped CDF has a simple form.

Lemma 11.5 (Laplace transform of F). *For $\Re s > 0$,*

$$\int_0^\infty e^{-sx} F(x) dx = \frac{G(-s)}{s}. \quad (158)$$

Proof. Stieltjes integration by parts on $[0, 1]$ gives

$$G(-s) = \int_0^1 e^{-sx} dF(x) = e^{-s} + s \int_0^1 e^{-sx} F(x) dx.$$

Since $F(x) = 1$ for $x \geq 1$, $\int_1^\infty e^{-sx} dx = e^{-s}/s$. Adding this tail proves (158). $\square$

Fourier inversion of an exponentially tilted F gives a Bromwich formula without any contour-shifting hypothesis. For every $r > 0$,

$$F(x) = e^{rx} \int_{\mathbb{R}} e^{2\pi i \xi x} \frac{G(-r - 2\pi i \xi)}{r + 2\pi i \xi} d\xi. \quad (159)$$

Set $r = 2^\lambda$ and use $rx = \lambda$. After the change of variable $\xi = rv/(2\pi\sqrt{\lambda})$, define the normalized saddle mass

$$\mathcal{S}(x) = \frac{F(x)\sqrt{2\pi\lambda(x)}}{\exp(r(x)x + q(r(x)))}. \quad (160)$$

Then exactly

$$\mathcal{S}(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} K_{\lambda(x)}(v) dv, \quad (161)$$

where, using the analytic logarithm of $G(-z)$ in $\Re z > 0$,

$$K_\lambda(v) = \frac{1}{1 + iv/\sqrt{\lambda}} \exp\left(i\sqrt{\lambda}v + q_{\mathbb{C}}(r(1 + iv/\sqrt{\lambda})) - q(r)\right). \quad (162)$$

The real quantity $\mathcal{S}(x)$ is positive because $F(x) > 0$.

Taking logarithms gives the exact identity

$$\log F(x) = rx + q(r) - \frac{1}{2} \log(2\pi\lambda) + \log \mathcal{S}(x). \quad (163)$$

No asymptotic approximation has yet been made.

11.5 The corrected sharp main term

Define

$$C = \frac{C_{\Gamma\zeta}}{L} - \frac{7L}{12} - \frac{1}{2} \log \pi \approx -2.027983898638859424, \quad (164)$$

and

$$\mathcal{M}(x) = -\frac{L}{2}\lambda(x)^2 + \left(1 + \frac{L}{2}\right)\lambda(x) - \frac{1}{2} \log \lambda(x) + C + \Psi(\lambda(x)). \quad (165)$$

Proposition 11.6 (Exact separation of the main term). *For small positive x , with $r = 2^{\lambda(x)}$,*

$$rx + q(r) - \frac{1}{2} \log(2\pi\lambda) = \mathcal{M}(x) + E(r). \quad (166)$$

Consequently

$$\log F(x) - \mathcal{M}(x) = \log S(x) + E(r). \quad (167)$$

Proof. Use $rx = \lambda$, $\log r = L\lambda$, and the exact decomposition (141):

$$q(r) = -\frac{L}{2}\lambda^2 + \frac{L}{2}\lambda + \bar{R} + \Psi(\lambda) + E(r).$$

Now

$$\bar{R} - \frac{1}{2} \log(2\pi) = \frac{C_{\Gamma\zeta}}{L} - \frac{L}{12} - \frac{1}{2} \log 2 - \frac{1}{2} \log \pi = C,$$

because $(1/2) \log 2 = L/2$. Collecting terms proves (166); combine it with (163) to obtain (167). $\square$

The analytic heart of the saddle argument is the following quantitative estimate.

Theorem 11.7 (Normalized saddle mass). *As $x \downarrow 0$,*

$$S(x) = 1 + O(\lambda(x)^{-1}). \quad (168)$$

Proof structure. Put $\lambda = \lambda(x)$. On a central arc $|v| \leq \lambda^\delta$ with a fixed small $\delta > 0$, expand the analytic logarithm in (162). The identities $rx = \lambda$ and the explicit derivatives of the periodic Laplace decomposition cancel the linear term and normalize the quadratic term, giving

$$\log K_\lambda(v) = -\frac{v^2}{2} + O\left(\frac{1 + |v|^3}{\sqrt{\lambda}}\right). \quad (169)$$

A one-order-finer expansion shows that the term of order $\lambda^{-1/2}$ is odd and purely imaginary, so its Gaussian integral vanishes. The remaining central error is $O(\lambda^{-1})$ after integration.

For the complementary arc, the infinite product (133) gives uniform vertical decay. Factors whose scale is comparable to r provide a strict modulus loss away from the real axis; additional factors give arbitrary polynomial decay. This produces an integrable majorant and makes the tail $o(\lambda^{-N})$ for the order needed here. Thus (161) equals the normalized Gaussian integral $(2\pi)^{-1/2} \int e^{-v^2/2} dv = 1$ plus $O(\lambda^{-1})$. $\square$

Theorem 11.8 (Sharp endpoint asymptotic). *As $x \downarrow 0$,*

$$\log F(x) = \mathcal{M}(x) + O(\lambda(x)^{-1}). \quad (170)$$

Equivalently,

$$F(x) = \frac{1}{\sqrt{\lambda(x)}} \exp\left(-\frac{L}{2}\lambda(x)^2 + \left(1 + \frac{L}{2}\right)\lambda(x) + C + \Psi(\lambda(x))\right) \left(1 + O(\lambda(x)^{-1})\right). \quad (171)$$

Since $\lambda(x) \asymp -\log x$, the logarithmic error is also $O(1/(-\log x))$.

Proof. By (167), the error is $\log \mathcal{S}(x) + E(r)$. The estimate (168) and $\log(1+u) = u + O(u^2)$ give $\log \mathcal{S} = O(\lambda^{-1})$. The forward tail is $E(2^\lambda) = O(e^{-2^\lambda})$, hence smaller than every inverse power of λ . This proves (170); exponentiation gives (171). $\square$

11.6 Compact Lambert-W form

Let

$$W = W_{-1}(-Lx), \quad \lambda = -W/L. \quad (172)$$

The sharp main term (165) is identically equal to

$$\mathcal{M}_W(x) = -\frac{7L}{12} - \frac{1}{2} \log(\pi x) + \frac{C_{\Gamma\zeta} - W - \frac{1}{2}W^2}{L} + \Psi\left(-\frac{W}{L}\right). \quad (173)$$

Indeed, $\log x = \log \lambda - L\lambda$ and $W = -L\lambda$ reduce (173) to (165). Therefore

$$\log F(x) = \mathcal{M}_W(x) + O\left(\frac{1}{-\log x}\right). \quad (174)$$

This is the shortest correct version of the leading asymptotic.

11.7 Expansion in elementary logarithms

Some applications prefer not to leave Lambert W in the answer. Put

$$T = -\log x, \quad H = \log T, \quad a = \log L. \quad (175)$$

Define

$$C_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6a^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi. \quad (176)$$

Then set

$$\begin{aligned} \mathcal{M}_{\text{elem}}(x) = & -\frac{T^2}{2L} - \frac{TH}{L} + \left(\frac{1}{2} + \frac{1+a}{L}\right)T \\ & - \frac{H^2}{2L} + \frac{aH}{L} + C_0 - \frac{H^2}{2LT} + \frac{aH}{LT}. \end{aligned} \quad (177)$$

Theorem 11.9 (Elementary-logarithm form). As $x \downarrow 0$,

$$\log F(x) = \mathcal{M}_{\text{elem}}(x) + \Psi(\lambda(x)) + O(T^{-1}). \quad (178)$$

Proof. Insert the Lambert expansion (155) into the nonoscillatory part of (165), retain all terms that are not $o(T^{-1})$, and simplify. This gives exactly (177). The saddle error is already $O(T^{-1})$. Crucially, the periodic term is *not* expanded by replacing λ with a coarse approximation: an $o(1)$ phase error can still change a nonconstant periodic function by an amount larger than the desired remainder. It must remain sampled at the exact lower-Lambert phase, or the phase itself must be expanded to matching precision and then propagated through the derivatives of Ψ . $\square$

For the dyadic scale $x = 2^{-t}$, a convenient nonoscillatory main term is

$$\begin{aligned} \mathcal{M}_{\text{dyad}}(t) = & -\frac{L}{2}t^2 - t \log t + \left(1 + \frac{L}{2}\right)t - \frac{(\log t)^2}{2L} + C \\ & - \frac{(\log t)^2}{2L^2t}. \end{aligned} \quad (179)$$

Then

$$\log F(2^{-t}) = \mathcal{M}_{\text{dyad}}(t) + \Psi(\lambda_t) + O(t^{-1}). \quad (180)$$

Substituting $T = Lt$ into (177) differs from (179) by only $(\log L)^2/(2L^2t)$, already within the stated error.

11.8 Why the periodic term cannot be omitted

A common but genuine asymptotic error

A formula of the form

$$\log F(x) = \mathcal{M}_{\text{elem}}(x) + O\left(\frac{1}{-\log x}\right)$$

with no periodic correction is false. The missing term is tiny numerically, but it is of order 1, not of order $1/\log x$.

Proposition 11.10 (Obstruction to a nonoscillatory remainder). *Let $M_0(x)$ be any main term satisfying $M_0(x) = \mathcal{M}(x) - \Psi(\lambda(x)) + o(1)$. Then*

$$\log F(x) - M_0(x) \not\rightarrow 0 \quad (181)$$

unless Ψ is identically zero.

Proof. For a fixed $u \in [0, 1)$, choose

$$\lambda_n = n + u, \quad x_n = \lambda_n 2^{-\lambda_n}. \quad (182)$$

Then $x_n \downarrow 0$ and the exact Lambert phase of x_n is λ_n . The sharp theorem gives

$$\log F(x_n) - M_0(x_n) = \Psi(n + u) + o(1) = \Psi(u) + o(1).$$

If the left side tended to zero for all such sequences, then $\Psi(u) = 0$ for every u . This contradicts the nonzero Fourier coefficient (148). $\square$

11.9 The first saddle correction

The normalized saddle mass has a full asymptotic expansion. Its first nontrivial term can be written explicitly using the derivatives of Ψ . Define

$$A(u) = \frac{\Psi'(u)}{L} - \frac{1}{2}, \quad (183)$$

$$B(u) = -\frac{1}{4} + \frac{1}{2L} + \frac{\Psi'(u)}{2L} - \frac{\Psi''(u)}{2L^2}. \quad (184)$$

On the central scale, the exponent in the saddle kernel has the form

$$\log K_\lambda(v) = -\frac{v^2}{2} + \lambda^{-1/2} P_1(v; u) + \lambda^{-1} P_2(v; u) + O(\lambda^{-3/2}(1 + |v|^9)), \quad (185)$$

where $u = \lambda$ modulo 1 and

$$P_1(v; u) = i \left(A(u)v + \frac{v^3}{3} \right), \quad P_2(v; u) = \frac{v^4}{4} + B(u)v^2. \quad (186)$$

The $\lambda^{-1/2}$ contribution integrates to zero by oddness. At order λ^{-1} , let Z be standard normal. The relative mass coefficient is

$$\begin{aligned} m_1(u) &= \mathbb{E} \left[P_2(Z; u) + \frac{1}{2} P_1(Z; u)^2 \right] \\ &= \frac{3}{4} + B(u) - \frac{1}{2} \left(A(u)^2 + 2A(u) + \frac{5}{3} \right). \end{aligned} \quad (187)$$

Using $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, and $\mathbb{E}Z^6 = 15$, simplification gives

$$\alpha_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + (\Psi'(u))^2}{2L^2}. \quad (188)$$

Theorem 11.11 (First corrected saddle term). *As $x \downarrow 0$,*

$$\log F(x) = \mathcal{M}(x) + \frac{\alpha_1(\lambda(x))}{\lambda(x)} + O(\lambda(x)^{-2}). \quad (189)$$

The coefficient α_1 is smooth and one-periodic.

Proof. Exponentiate (185), integrate termwise against the normalized Gaussian, and use (187). This gives $\mathcal{S}(x) = 1 + \alpha_1(\lambda)/\lambda + O(\lambda^{-2})$. Taking the logarithm does not change the first coefficient. The forward tail remains exponentially small, so (167) proves the result. $\square$

11.10 The all-orders expansion

Theorem 11.12 (Full Poincare expansion). *There exist smooth one-periodic functions $\alpha_j : \mathbb{R} \rightarrow \mathbb{R}$ for $j \geq 1$ such that, for every integer $N \geq 1$,*

$$\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{\alpha_j(\lambda(x))}{\lambda(x)^j} + O(\lambda(x)^{-N}) \quad (x \downarrow 0). \quad (190)$$

The first coefficient is (188). Every α_j is obtained by a finite algebraic-differential expression in Ψ and its derivatives, with coefficients in $\mathbb{Q}(L)$.

Construction and proof. Expand the logarithm of the scaled kernel to arbitrary order:

$$\log K_\lambda(v) = -\frac{v^2}{2} + \sum_{m=1}^{2N} \lambda^{-m/2} P_m(v; \lambda) + \mathcal{R}_N(v, \lambda). \quad (191)$$

Each P_m is a polynomial in v whose coefficients are smooth periodic combinations of derivatives of Ψ . Uniform derivative bounds follow from the absolutely convergent Fourier series (146). Taylor's theorem gives a central-arc bound on $\mathcal{R}_N$; the vertical product estimates make the complementary arc smaller than λ^{-N} .

Exponentiating (191) produces complete Bell polynomials in $P_1, P_2, \dots$. Integrating termwise against $e^{-v^2/2}$ replaces every monomial by a Gaussian moment. Odd half-integer powers vanish by parity, leaving

$$\mathcal{S}(x) = 1 + \sum_{j=1}^{N-1} \frac{m_j(\lambda(x))}{\lambda(x)^j} + O(\lambda^{-N}), \quad (192)$$

with smooth periodic m_j .

Finally, write $1 + \sum_{j \geq 1} m_j z^j = \exp(\sum_{j \geq 1} \alpha_j z^j)$. Comparing derivatives gives the finite recursion

$$\alpha_n = m_n - \frac{1}{n} \sum_{k=1}^{n-1} k \alpha_k m_{n-k}, \quad n \geq 1. \quad (193)$$

This defines the periodic logarithmic coefficients. Insert the resulting expansion of $\log \mathcal{S}$ into (167); the forward tail $E(2^\lambda)$ is beyond all algebraic orders and does not affect any coefficient. $\square$

Remark 11.13 (Two independent all-orders operations). There are two distinct expansions:

1. the saddle expansion in powers of $1/\lambda$, whose coefficients are periodic;
2. the Lambert-phase expansion of λ in powers of $1/(-\log x)$ with polynomial logarithmic coefficients.

They can be composed, but the periodic phase must be propagated to the same order. Blindly replacing λ by the first two elementary terms inside $\Psi(\lambda)$ loses control of the claimed remainder.

12 Synthesis and computational workflow

The theory is unusually coherent: each viewpoint supplies the difficult step for a different class of questions.

Viewpoint	What it makes transparent
Probability	Existence, positivity, symmetry, support, weak convergence, the negative Laplace product.
Infinite convolution	Smoothness, constructive approximation, Fourier product, rapid decay.
Delay equation	Self-similarity, endpoint flatness, derivative identities.
Thue–Morse extension	A global dilation equation, exact dyadic derivatives, closed rational formulas.
Moments	Rational recurrences, inverse-power values, denominator control.
Taylor blocks	Fast exact evaluation following the 1-bits of the numerator.
Poisson summation	Partition of unity, sampling identities, cosine expansion.
Legendre analysis	Exact orthogonal coefficients and optimal polynomial approximation.
Negative Laplace analysis	Log-periodic structure and the Gamma–zeta correction.
Lambert saddle analysis	The sharp endpoint expansion and all higher corrections.

For exact arithmetic at a dyadic $a/2^N$, the recommended workflow is:

1. generate $d_0, \dots, d_N$ by (68);
2. use [theorem 7.5](#) to remove highest binary digits;
3. use the closed formula (80) independently when a denominator certificate or a cross-check is desired;
4. derive exact derivatives from (92);
5. reduce the resulting rational only at controlled points if expression swell matters.

For tiny non-dyadic x , the sharp asymptotic should be evaluated in this order:

1. compute the lower-branch phase $\lambda = -W_{-1}(-x \log 2)/\log 2$;
2. compute the rapidly convergent Fourier series for $\Psi(\lambda)$, normally using only its first harmonic at ordinary precision;
3. evaluate (165) or (173);
4. add $\alpha_1(\lambda)/\lambda$ if a second-order logarithmic approximation is needed.

The exact phase is cheap compared with the enormous dynamic range of $F(x)$, and retaining it avoids the only structurally serious error in the naive elementary formula.

13 Corrections and strengthened statements

A formal development is particularly useful for this topic because many formulas are correct in spirit but sensitive to endpoint conventions, global-versus-bounded notation, or a single power of two. The development underlying this article makes the following points explicit.

1. **Bounded versus signed Fabius functions.** The clamped $F: \mathbb{R} \rightarrow [0, 1]$ does not satisfy $F'(x) = 2F(2x)$ for all positive x ; the signed extension Θ does. Every global derivative formula in this article is stated for Θ , then restricted to $[0, 1]$ when appropriate.
2. **Finite convolution approximants.** At a finite stage one must convolve only the finite family of uniform kernels that has actually been introduced. An infinite-convolution notation at a finite step obscures the induction and is not literally correct.
3. **Endpoint weights in step approximants.** The indicator of a closed dyadic cell must carry half-weight at its endpoints if the stated pointwise convergence is intended to include grid boundaries. The interior convergence is unaffected, but the literal endpoint theorem otherwise fails.
4. **Poisson summation.** The translated formula is

$$\sum_m \text{up}(t + am) = a^{-1} \sum_k \widehat{\text{up}}(k/a) e^{2\pi i k t / a};$$

the translation parameter and exponential factor cannot be omitted. Statements with a positive integer subdivision parameter require that parameter to be strictly positive.

5. **Removable singularities.** The factor $(e^z - 1)/z$ in (67) is an entire function after assigning the value 1 at $z = 0$. Coefficient arguments should use this completed function rather than silently divide by zero.
6. **Taylor-remainder hypotheses.** The translated-remainder lemma behind theorem 7.3 needs the scale and derivative order to stay in the range where the iterated dilation formula is valid. The formal theorem records the missing inequality instead of relying on an implicit convention about negative indices.
7. **The sign of the Reshetnikov exponent.** The definition is $2^{\binom{n-1}{2}} (2n)! F(2^{-n}) \cdots$, with a positive exponent as in (99). A negative exponent would contradict the integral values already at small indices.
8. **Normalization of the closed dyadic formula.** The powers of two in the Taylor-integral change of variables must be tracked through both the polynomial factor and the differential. The normalized form is (80); it is stable under dyadic refinement and agrees with the exact evaluator.

9. **The sharp asymptotic needs a periodic term.** The nonconstant function $\Psi(\lambda(x))$ in (170) cannot be absorbed into an $O(1/(-\log x))$ remainder. The subsequence argument in [theorem 11.10](#) gives a direct disproof of the uncorrected claim.
10. **All-orders expansion requires exact phase bookkeeping.** The saddle coefficients are periodic functions of the lower-Lambert phase. A truncated expansion of λ may be substituted only after its error has been propagated through the derivatives of those periodic coefficients.

A Correspondence with the Lean development

The proof organization above follows the dependency graph of the Lean source rather than the order of the historical papers. The table lists representative modules; several technical bridge files are omitted from the right column for readability.

Mathematical topic	Representative Lean modules
Bounded specification, regularity, and the F -up bridge	<code>Basic.lean</code> , <code>OriginalPaperSupplement.lean</code>
Probability, moments, and generating functions	<code>AnalyticMoments.lean</code> , <code>Paper05442.lean</code>
Closed dyadic values and the exact evaluator	<code>DyadicClosedForm.lean</code> , <code>TaylorReduction.lean</code>
Arithmetic sequences and denominator results	<code>Arithmetic.lean</code> , <code>PaperStatements.lean</code>
Finite q -binomial and Thue–Morse identities	<code>FabiusQBinomialFormula.lean</code> ; <code>FabiusDyadicQBinomialScalar.lean</code>
Legendre coefficients and approximation	<code>FabiusLegendreCoefficients.lean</code>
Negative Laplace product and periodic correction	<code>NegativeLaplace.lean</code> ; <code>PeriodicCorrection.lean</code> ; <code>PeriodicMean.lean</code>
Lower Lambert phase and explicit saddle radius	<code>FabiusLambertPhase.lean</code> , <code>FabiusLambertSaddle.lean</code>
Bromwich inversion and exact saddle reduction	<code>FabiusBromwichInput.lean</code> ; <code>FabiusSaddleReduction.lean</code> ; <code>FabiusSharpExactReduction.lean</code>
Sharp main term and elementary logarithmic form	<code>FabiusSharpConstant.lean</code> ; <code>FabiusSharpLambertMain.lean</code> ; <code>FabiusWikipediaMain.lean</code> ; <code>FabiusWikipediaExpansion.lean</code>
First and all-orders saddle corrections	<code>FabiusFirstSaddleCorrection.lean</code> ; <code>FabiusFullAsymptoticExpansion.lean</code>
Coverage and audit	Repository-level <code>README.md</code> ; <code>PAPER_COVERAGE.md</code> ; <code>ASYMPTOTIC_COMPLETION_AUDIT.md</code>

The formal source separates statements over $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Q}$ more sharply than an ordinary exposition does. In particular, the exact evaluator lives over the rationals and is later cast to the reals; the analytic uniqueness theorem then shows that its real value is independent of the chosen representative satisfying the Fabius axioms. This separation is what turns “the algorithm appears to work” into a theorem about the unique mathematical function.

B Selected exact data

For convenient checking, here are several sequences in aligned form.

Moments

$$(c_n)_{n \geq 0} = 1, \frac{1}{9}, \frac{19}{675}, \frac{583}{59535}, \frac{132809}{32531625}, \frac{46840699}{24405225075}, \dots,$$

$$(d_n)_{n \geq 0} = 1, \frac{1}{2}, \frac{5}{18}, \frac{1}{6}, \frac{143}{1350}, \frac{19}{270}, \frac{1153}{23814}, \frac{583}{17010}, \dots$$

Inverse-power values

$$F(1) = 1, \quad F(1/2) = \frac{1}{2}, \quad F(1/4) = \frac{5}{72},$$

$$F(1/8) = \frac{1}{288}, \quad F(1/16) = \frac{143}{2\,073\,600}, \quad F(1/32) = \frac{19}{33\,177\,600},$$

$$F(1/64) = \frac{1153}{561\,842\,749\,440}, \quad F(1/128) = \frac{583}{179\,789\,679\,820\,800}.$$

Asymptotic constants

$$L = 0.6931471805599453094\dots,$$

$$C_{\Gamma\zeta} = -0.7286939170039306059\dots,$$

$$\bar{R} = -1.1090453654341866822\dots,$$

$$C = -2.0279838986388594240\dots$$

The magnitude of the first periodic Fourier coefficient is $1.0627109779221651 \times 10^{-6}$; the second is already about 6.69×10^{-13} .

C Notation index

Symbol	Meaning
F	Bounded, clamped Fabius CDF, equal to 0 on $(-\infty, 0]$ and 1 on $[1, \infty)$.
up	Rvachev's even C^∞ bump supported on $[-1, 1]$.
Θ	Signed Thue–Morse global extension of F .
$w_2(n)$	Number of 1-bits in n .
ε_n	Thue–Morse sign $(-1)^{w_2(n)}$.
c_n	Even moment $\int x^{2n} \text{up}(x) dx$.
d_n	Half-moment $n \int_0^1 x^{n-1} \text{up}(x) dx$, with $d_0 = 1$.
F_n, G_n	Integral numerator sequences associated with c_n, d_n .
R_n	Reshetnikov number in (99).
D_n	Common denominator of reduced odd-grid values at level n .
$G(z)$	Moment generating function $\mathbb{E}(e^{zX})$.
$q(s)$	$\log G(-s)$ for $s > 0$.
$R(t)$	Exact one-periodic correction in the negative Laplace logarithm.
$\Psi(t)$	Centered periodic correction $R(t) - \bar{R}$.

$\lambda(x)$	Lower-Lambert phase solving $\lambda 2^{-\lambda} = x$.
$\mathcal{S}(x)$	Normalized saddle mass in (160).
$\mathcal{M}(x)$	Corrected sharp logarithmic main term in (165).

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